

Chapter 7: Cosmological Implications and Gravitational Equivalence

7.1 Reconciling Quantum Resonance with Spacetime

The BMI framework reconciles quantum mechanics and General Relativity by revealing that curvature is the long-range manifestation of **Bridge Stress** (τ) and **Inter-brane Potential** (V_{gap}) generated by the collision of branes σ_A and σ_B . Spacetime is not a passive fabric; it is an active bimodal medium whose tension decays radially from localized harmonic modes.

7.2 Gravity as Global Metric Equilibrium (Δ)

Gravitational acceleration is the drift effect of the system seeking the global minimum of the **Metric Equilibrium** (Δ). We derive the local curvature as a function of the combined tension of the colliding branes:

$$g_{\mu\nu}(t) = \eta_{\mu\nu} + \Xi_{\mu\nu}(\tau(t) + V_{\text{gap}}(t))$$

Gravity is the natural minimization of this collective potential energy. Matter moves toward matter because the bimodal medium experiences an optimization of its elastic deformation energy.

7.3 Higher-Dimensional Toroidal Boundaries (T^n)

While particles stabilize within T^3 , higher-dimensional degrees of freedom (T^n) are required to maintain global topological continuity. These dimensions represent the degrees of freedom that constrain the frequency limits of the harmonic resonance spectrum, preventing the infinite divergences found in traditional quantum gravity.

7.4 Dark Energy as Inter-brane Potential

Cosmic expansion is the continuous, global phase-relaxation of the bimodal field toward **Asymptotic Flatness**. "Dark Energy" is identified as the residual **Inter-brane Potential**

(V_{gap}). As the branes coalesce, this potential energy is converted into the kinetic drive of the expansion. This relaxation path explains the accelerating velocity curve, linking the macroscopic observation of Dark Energy directly to the underlying quantum topology.